

Arithmophysics: Colors Inside the Subpower Bound

Signed boundary energy, complete prime packets, and record descent

Christophe Michaels research programme

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Directive: investigate the arithmetic structure that controls growth

Continue from the proved orthogonal boundary compression. Investigate Christophe Michaels's proposal that prime colors expose the cancellation responsible for subpower growth. Give colors their recovered arithmetic meaning: prime valuation coordinates, squarefree support, factor-count parity, and mixed-prime interactions.

The structural approximation is already controlled uniformly. Focus on the arithmetic size of the retained energy. Start with the simplest exact identities, try a direct comparison, and then combine color information with completed prime packets and the existing record-descent theorem. A weaker estimate that closes through descent is sufficient.

Execute the mathematics. State each independent input and carry it through the original metric. An equivalent sufficient condition is a legitimate target; its proof must obtain additional control from arithmetic. Keep failures specific to the estimates tested and preserve successful partial results.

1. The fixed target and the completed round trip

Set

$$M(x) = \sum_{n \leq x} \mu(n), \quad c_N(n) = \frac{\mu(n)}{n}, \quad B_N(a, b) = \min(a, b).$$

The complete energy is

$$R(N) = c_N^* B_N c_N = \sum_{k < N} \frac{M(k)^2}{k(k+1)} + \frac{M(N)^2}{N}. \quad (1)$$

The terminal is part of this quantity.

Use the mesh from *Orthogonal Boundary Compression*:

$$d = N^{-1/6}, \quad x_0 = 0, \quad x_{r+1} = \min\{N, x_r + \max(1, \lfloor dx_r^{2/3} \rfloor)\}.$$

It ends at $x_m = N$, with $m \leq 13\sqrt{N} + 2$. On $(a, b] = (x_{r-1}, x_r]$, let

$$Q_r = \sum_{a < n \leq b} \mu(n), \quad H_r = \sum_{a < n \leq b} \frac{\mu(n)}{n}.$$

For $a > 0$, the interval contributes boundary coefficients

$$p_a^{(r)} = \frac{bH_r - Q_r}{b - a}, \quad p_b^{(r)} = \frac{Q_r - aH_r}{b - a}. \quad (2)$$

The initial singleton is unchanged; add contributions at shared boundaries. Write this linear projection as $p = \mathcal{T}_N c_N$, and $z = c_N - p$. It is orthogonal in the B_N metric:

$$R(N) = E_{\mathcal{P}}(N) + E_{\text{res}}(N), \quad E_{\mathcal{P}} = p^* B_N p, \quad 0 \leq E_{\text{res}} \leq C_0 := 2^{-4/3}. \quad (3)$$

The mesh is independent of the signs and of the spectral cutoff. The same bound holds for every real physical source $|a_n| \leq 1$, with coefficients a_n/n .

For

$$u_N(x) = \sum_{n \leq N} \frac{\mu(n)}{n} \min(x, n),$$

the retained energy has the exact forms

$$E_{\mathcal{P}}(N) = \sum_r \frac{[u_N(x_r) - u_N(x_{r-1})]^2}{x_r - x_{r-1}}, \quad E_{\mathcal{P}}(N) \geq \frac{M(N)^2}{N}. \quad (4)$$

In each interval the mean slope is

$$\frac{u_N(b) - u_N(a)}{b - a} = \sum_{n > b} \frac{\mu(n)}{n} + \frac{Q_r - aH_r}{b - a}. \quad (5)$$

Keep the inherited tail and the local contribution together inside the square.

The main goal is

$$E_{\mathcal{P}}(N) \leq C_\varepsilon N^\varepsilon \quad \text{for every } \varepsilon > 0. \quad (\text{Goal})$$

Equation (3) gives the complete return at an additive cost C_0 . The supplied paper also provides the alternative low-mode return if that route yields a better estimate.

2. First route: an exact colored Gram matrix in the original metric

The recovered color coordinate is

$$\nu(n) = (\nu_p(n))_p, \quad \omega(n) = \#\{p : p \mid n\}, \quad \Omega(n) = \sum_p \nu_p(n).$$

The immediate algebraic handle is

$$\mu(n) = \mu(n)^2 (-1)^{\omega(n)}. \quad (6)$$

For $k \geq 0$, define the nonnegative color source and its retained projection by

$$c_N^{[k]}(n) = \frac{\mu(n)^2}{n} \mathbf{1}_{\{\omega(n)=k\}}, \quad p_k = \mathcal{T}_N c_N^{[k]}.$$

All colors use the same mesh and the same linear projection. Then

$$p = \sum_k (-1)^k p_k, \quad \mathbf{G}_{k\ell}(N) = p_k^* B_N p_\ell, \quad E_{\mathcal{P}}(N) = \sum_{k,\ell} (-1)^{k+\ell} \mathbf{G}_{k\ell}(N). \quad (7)$$

This is a concrete meaning of looking for colors inside the subpower bound.

First calculate this matrix. Identify which mixed-color entries suppress the large coherent contribution in the parity direction. Then refine the grading by selected small-prime incidence and the factor count outside that prime set. This distinguishes dependence on particular primes from dependence only on the number of factors.

Use the corresponding vector polynomial

$$\mathcal{F}_N(\zeta) = \sum_k \zeta^k p_k.$$

It satisfies

$$E_{\mathcal{P}} = \|\mathcal{F}_N(-1)\|_{B_N}^2, \quad \frac{1}{2\pi} \int_0^{2\pi} \|\mathcal{F}_N(e^{it})\|_{B_N}^2 dt = \sum_k G_{kk}. \quad (8)$$

Seek an arithmetic estimate at the actual parity phase $\zeta = -1$. Derive the cost of any passage from an angular average to this phase.

Concrete candidates to attempt:

1. A factorization or completion of squares that isolates a large coherent component annihilated, exactly or quantitatively, by the parity vector.
2. A bound for the complete mixed-color correction against a diagonal or smaller-horizon budget.
3. A finite-difference estimate across adjacent factor counts, derived from prime insertion with its inherited cutoffs.
4. A bound only on the record configurations in Section 4, if that is more accessible than a uniform estimate at every N .

Positive semidefiniteness of G supplies a lower bound. The needed result is an upper bound in its specific arithmetic parity direction. Preserve all mixed colors in that calculation.

3. Second route: complete prime-color packets before estimating

Let $\mathcal{C}$ be a finite prime set, $q_{\mathcal{C}} = \prod_{p \in \mathcal{C}} p$, and

$$f_{\mathcal{C},N}(m) = \frac{\mu(m)}{m} \mathbf{1}_{\{(m, q_{\mathcal{C}})=1\}}.$$

On the finite coefficient space, define

$$\mathcal{D}_{p,N} e_m = \begin{cases} p^{-1} e_{pm}, & pm \leq N, \\ 0, & pm > N. \end{cases}$$

Unique factorization gives the exact identity

$$c_N = \prod_{p \in \mathcal{C}} (I - \mathcal{D}_{p,N}) f_{\mathcal{C},N}. \quad (9)$$

Verify it and use it on completed packets. Begin with $\{2, 3\}$ and $\{2, 3, 5\}$, then choose any enlargement from the resulting calculation.

Apply $\mathcal{T}_N$ to the complete expression and evaluate its signed B_N -energy. Establish the cancellation supplied by the mixed prime differences together with the coprime source. Every floor cutoff, truncated descendant, and inherited boundary contribution stays in the expression.

Compute the relevant operator or arithmetic-family bound before multiplying separate norms. In particular, source truncation in the B_N metric requires its own accounting. The exact dilation isometry on a supported child does not automatically give a contraction for a truncation of an arbitrary parent.

A prior prime-by-prime induction had large intermediate low-mode populations. Use that as a calibration when testing a proposed invariant. The task here permits a gain from a complete signed packet, even when its intermediate factors grow.

Try to connect (9) to (7): determine whether the observed mixed-color cancellation can be written as a compulsory prime-packet response with a uniform estimate.

4. Third route: use colors to obtain a contracting smaller-horizon estimate

The earlier *Five-Level Architecture and Direct Energy Descent* contains an existing reduction that narrows the arithmetic task. Let

$$\mathcal{S}(X) = \max_{0 \leq n \leq X} R(n), \quad R(0) = 0, \quad \tau_h = \min\{n : |M(n)| = h\}.$$

Every strict energy record is an absolute-Mertens first passage. At a noninitial record N , write $h = |M(N)|$ and $a = \tau_{h-1}$. Keep a as the preceding adjacent-level first passage, whether or not it is itself an energy record.

The recovered adaptive reduction retains only records satisfying

$$N - a < \frac{\sqrt{N}}{[1 + \mathcal{S}(N^{1/6})]^{5/2}}, \quad (10)$$

together with the initial record. If $T_{\text{ret}}(X)$ is the maximum of 1 and $M(N)^2/N$ over these retained records through X , the proved accounting is

$$\mathcal{S}(X) \leq (4 \log X + 4)T_{\text{ret}}(X) + 2(\log X)^2[1 + \mathcal{S}(X^{1/6})]^5. \quad (11)$$

Thus the following estimate is sufficient only on the retained set:

$$\frac{M(N)^2}{N} \leq C_\varepsilon N^\varepsilon [1 + \mathcal{S}(N^{1/6})]^5. \quad (12)$$

A bound of the same size on $E_{\mathcal{P}}(N)$ there also suffices by (4). Try to prove (12), or its boundary-energy version, using the actual color interactions. This permits a weaker estimate than the direct Goal at every horizon.

More generally, a proved recurrence

$$1 + \mathcal{S}(X) \leq C_\varepsilon X^\varepsilon [1 + \mathcal{S}(X^\beta)]^q, \quad q\beta < 1, \quad (13)$$

for every $\varepsilon > 0$, closes the argument. Indeed, the elementary bound $\mathcal{S}(X) \leq 2X$ makes

$$\rho = \limsup_{X \rightarrow \infty} \frac{\log(1 + \mathcal{S}(X))}{\log X}$$

finite, and (13) gives $\rho \leq \varepsilon + q\beta\rho$, hence $\rho = 0$.

There is a concrete finite reconstruction to examine. Put $K = \lfloor N^{1/6} \rfloor$ and $A = \mu \mathbf{1}_{[1, K]}$. With Dirichlet convolution and $\mathbf{1}(n) = 1$,

$$\mu(n) = \sum_{j=1}^6 (-1)^{j-1} \binom{6}{j} (A^{*j} * \mathbf{1}^{*(j-1)})(n), \quad n \leq N. \quad (14)$$

The six terms are one complete signed expression. Compute their joint boundary energy and mixed-color terms. Investigate whether this supplies an exponent $q < 6$ at scale $N^{1/6}$, or another pair with $q\beta < 1$. The number of factors in (14) is not itself an energy estimate; establish the quantitative bound with its full multiplicity and projection costs.

5. Fourth route: signed work, telescoping, and an arithmetic correction

Set

$$D_N = \sum_{n \leq N} \frac{\mu(n)^2}{n} \leq 1 + \log N.$$

The exact work identity and compressed deficit are

$$R(N) - D_N = 2 \sum_{n=2}^N \frac{\mu(n)M(n-1)}{n}, \quad E_{\mathcal{P}}(N) - D_N = 2 \sum_{n=2}^N \frac{\mu(n)M(n-1)}{n} - E_{\text{res}}(N). \quad (15)$$

The trial inequality $E_{\mathcal{P}} \leq D_N$ is sufficient and stronger than the main Goal. Attempt it, but retain the more flexible target $E_{\mathcal{P}} \leq C_{\varepsilon} N^{\varepsilon}$, or the record-descent target, if that specific comparison fails.

Search for a correction built from prime colors, complete return blocks, or divisor packets that converts the signed work into a controlled boundary term and a nonnegative remainder. Specify the correction explicitly and prove its endpoint bound independently.

For scale comparisons, preserve the exact prefix–tail formula:

$$R(N) = R(L) + 2M(L) \sum_{L < n \leq N} \frac{\mu(n)}{n} + \sum_{L < a, b \leq N} \frac{\mu(a)\mu(b)}{\max(a, b)}. \quad (16)$$

Use it to test a growth recurrence with the cross term intact. A bound on each positive term separately is useful only if the resulting complete recurrence closes.

A finite Gram or Schur calculation can guide the search for a certificate. Derive the uniform arithmetic certificate from its entries; finite matrix fitting supplies a candidate to prove.

6. Reuse the earlier positive color carriers with an explicit observation cost

The recovered color bridge supplies the finite-period functions

$$h_{p,a}(r) = \mathbf{1}_{\{p^a|r\}} - \frac{1}{p} \mathbf{1}_{\{p^{a-1}|r\}}, \quad h_n(r) = \frac{c_n^{\text{Ram}}(r)}{n}.$$

For a common period divisible by $\text{lcm}(1, \dots, N)$,

$$\langle h_m, h_n \rangle = \delta_{mn} \frac{\varphi(n)}{n^2}, \quad h_n(1) = \frac{\mu(n)}{n}. \quad (17)$$

It also supplies the Smith–Jordan divisor identity, for $s > 0$,

$$J_s(d) = d^s \prod_{p|d} (1 - p^{-s}), \quad \sum_{m,n} \overline{a_m} \frac{\gcd(m,n)^s}{(mn)^{s/2}} a_n = \sum_d J_s(d) \left| \sum_{d|n} \frac{a_n}{n^{s/2}} \right|^2. \quad (18)$$

These carriers are available if they produce a useful estimate for the actual boundary energy. Derive the finite observation map, its norm, and every localization remainder. In particular, (17) is orthogonality over a complete period, while the arithmetic source is observed at $r = 1$.

The gcd kernel in (18) and the original minimum kernel have different entries. Establish their comparison on the required arithmetic family, keeping any signed difference. For example, the matrix $\min(m, n) - \gcd(m, n)$ has zero diagonal and nonzero off-diagonal entries, so it cannot be treated as a positive remainder.

Prior results about a singular theta observation concern their stated family and metric. Test the present finite boundary map directly. Use these positive carriers as tools for (7), (12), or (15), with the return cost written down.

7. Calculations and controls that can distinguish the mechanisms

Reuse the already checked orthogonal-compression baseline:

$$N = 16384, \quad m = 382, \quad E_{\mathcal{P}} = 1.588343108, \quad E_{\text{res}} = 0.015097113.$$

Start with $N = 256, 1024, 4096, 16384$, adding larger horizons only to resolve a specific candidate's scaling or a suspected failure.

For the color calculation, record the Gram matrix, parity-direction energy, same-color contribution, mixed-color contribution, and refinements by small-prime signature. For the packet calculation, record the complete signed output and all inherited cutoffs. For descent, evaluate the proposed allowance specifically at the retained records.

Use three distinct controls with the same mesh:

1. Coherent positive charges, to check the source-independent geometry.
2. Independent random signs on squarefree support, to test which effects survive loss of multiplicative dependence.
3. Shuffled Möbius signs within prescribed size blocks on squarefree support, preserving the sign counts but disrupting their arithmetic locations.

The authentic parity-colored source is defined by (6). Apply control signs explicitly rather than assigning them Möbius parity by construction. State exactly what each shuffle or randomization preserves.

For a promising identity, compare the two sides exactly at small integer horizons or with certified error when necessary. Use the finite checks to verify formulas and select an analytic claim, then prove that claim with its quantifiers.

8. Execution order and required return

1. Establish the colored boundary Gram representation and examine the simplest candidate cancellation.
2. Test completed small-prime packets as the mechanism behind that cancellation.
3. Apply any partial estimate to the record-descent target before demanding a stronger uniform result.
4. Attempt the signed-work correction if it offers a more direct proof.
5. Use the older orthogonal color carriers only with their observation cost and original-metric return included.

The desired result is one sufficient independent arithmetic inequality. Combine routes when the terms match exactly.

Return:

- The strongest proved estimate and its full proof.
- The exact color or prime relation that creates the gain.
- Its constants, horizon range, cutoff dependence, and total return cost.
- Compact reproducible calculations supporting the algebra and proposed mechanism.
- A completed return through (3), or a proved descent with $q\beta < 1$, if obtained.
- Otherwise, the precise term that remains and a sharply specified next inequality justified by the calculation.

Do not end with a generic status statement. Explain what the actual calculation establishes. If a simpler argument closes the bound, give that proof first and complete its return.

Source checkpoint

The prompt carries forward the following retrieved work:

- *Michaels Arithmophysics: Orthogonal Boundary Compression*, 6 October 2026: two-moment projection, bounded full-energy error, sample energy, and terminal preservation. Its proof and five Möbius diagnostic rows were checked in this conversation.
- *Michaels Prime Color Dual Kernel Bridge*, 21 September 2026, Sections 1–8 and 12–14: valuation colors, parity, additive/tensor kernels, Jordan identities, Ramanujan recombination, and the scope of observation costs.
- *Prime DNA: Five-Level Architecture and Direct Energy Descent*, 5 September 2026, Sections 8–11: retained-record recovery, adaptive threshold, sufficient contraction, and finite six-term reconstruction. The arithmetic estimate (12) is the target of that reduction.
- *Michaels Theory Atlas: Purposes, Energy Work, and Connections*, 23 September 2026, entries E12, M03, M07–M08, M11–M12, and M15–M16: exact source work, original Green storage, color parity, completed prime response, and record structure.

Preserve the original arithmetic source and all proved approximation bounds. Write a new continuation with the successful estimate, its complete return, and a concise resume point.